

Monads are monadic and locally presentable: a double perspective

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Fibrant double categories

Definition

A **pseudo double category** $\mathbb{D}$ is a pseudo category object in $\mathbf{Cat}$.

Convention. All double categories will be pseudo.

Notation. $\mathbb{D}$ consists of a *category* $\mathbb{D}_0$ of objects and tight arrows, a *category* $\mathbb{D}_1$ of loose arrows and cells

$$\begin{array}{ccccc}
 X & \xrightarrow{A} & Y & & X & \xrightarrow{A} & Y & & X & \xrightarrow{1_X} & X \\
 f \downarrow & f \alpha^g & \downarrow g & & \text{id}_X \downarrow & 1_A & \downarrow \text{id}_Y & & f \downarrow & 1_f & \downarrow f \\
 Z & \xrightarrow{B} & W, & & X & \xrightarrow{A} & Y, & & Y & \xrightarrow{1_Y} & Y
 \end{array}$$

source, target, identity and composition *functors*

$$\begin{array}{c}
 \xleftarrow{s} \\
 \mathbb{D}_0 \xrightarrow{1} \mathbb{D}_1 \xleftarrow{\odot} \mathbb{D}_1 \times_{\mathbb{D}_0} \mathbb{D}_1, \\
 \xleftarrow{t}
 \end{array}$$

and natural isomorphisms α, ℓ, r (associator, left and right unitors) subject to the usual axioms and coherence conditions.

Definition

A double category $\mathbb{D}$ is called **fibrant** if $\langle s, t \rangle: \mathbb{D}_1 \rightarrow \mathbb{D}_0 \times \mathbb{D}_0$ is a fibration.

In a fibrant double category, we can turn tight arrows to loose arrows (their liftings through the fibration) thus obtaining a rich interplay between $\mathbb{D}_0$ and $\mathbb{D}_1$.

Notation. For a tight arrow $f: X \rightarrow Y$ we denote by $\hat{f}: X \rightarrowtail Y$ and $\check{f}: Y \rightarrowtail X$ its **companion** and **conjoint** respectively with

$$\begin{array}{ccccc}
 X & \xrightarrow{\hat{f}} & Y & X & \xlongequal{\quad} & X & Y & \xrightarrow{\check{f}} & X & X & \xlongequal{\quad} & X \\
 f \downarrow & p_1 & \parallel & \parallel & p_2 & \downarrow f & \parallel & q_1 & \downarrow f & f \downarrow & q_2 & \parallel \\
 Y & \xlongequal{\quad} & Y & X & \xrightarrow[\hat{f}]{} & Y & Y & \xlongequal{\quad} & Y & Y & \xrightarrow[\check{f}]{} & X
 \end{array}$$

as the appropriate cells.

Examples. 1. The double category $\text{Span}(\mathcal{C})$ of spans in a category $\mathcal{C}$ with pullbacks. $\text{Span}(\mathcal{C})$ consists of the objects, arrows, spans and span homomorphisms of $\mathcal{C}$. Composition is given by pullbacks. For a tight $f : X \rightarrow Y$, $\hat{f}$ is $X \xrightarrow{\text{id}_X} X \xrightarrow{f} Y$ and $\check{f}$ is $Y \xleftarrow{f} X \xrightarrow{\text{id}_X} X$.

2. The double category $\text{Rel}(\text{Set})$ of sets, functions, relations and inclusion of relations. Composition is composition of relations. For a function $f : X \rightarrow Y$, $\hat{f}$ is the graph of f and $\check{f}$ is the opposite relation f° .

3. The double category Ring of rings, ring homomorphisms, bimodules and bimodule homomorphisms. Composition is given by the tensor product. For a ring homomorphism $f : R \rightarrow S$, $\hat{f}$ is ${}_R S_S$ and $\check{f}$ is ${}_S S_R$.

4. For a monoidal $\mathcal{V}$ with small coproducts preserved by $\otimes$ in each variable, $\mathcal{V}\text{-Mat}$ is the double category of sets, functions, profunctors $Y^{\text{op}} \times X \rightarrow \mathcal{V}$ with X, Y sets (called $\mathcal{V}$ -**matrices**) and natural transformations. Composition of $T : Y \rightarrowtail Z$ and $S : X \rightarrowtail Y$ is given by $\left\{ \sum_{y \in Y} T_{z,y} \otimes S_{y,x} \right\}_{z,x}$. For any function $f : X \rightarrow Y$, $\hat{f}$ and $\check{f}$ are the family $\{I \text{ if } f(x) = y \text{ and } 0 \text{ otherwise}\}_{y,x}$.

Definition [ASV]

A *fibrant* double category will be called **locally presentable** if

- $\mathbb{D}_0, \mathbb{D}_1$ are locally presentable,
- $\mathfrak{s}, \mathfrak{t}$ are cocontinuous right adjoints and
- for every loose arrow A , the functors $- \odot A$ and $A \odot -$ are accessible.

Examples [ASV]

1. $\text{Span}(\mathcal{C})$ is locally presentable when $\mathcal{C}$ is locally presentable.
2. $\text{Rel}(\text{Set})$ is locally presentable.
3. $\mathcal{V}\text{-Mat}$ is locally presentable when $\mathcal{V}$ is locally presentable and $\otimes$ preserves filtered colimits in each variable.

(co)Monads

Fix a double category $\mathbb{D}$. A **monad** in $\mathbb{D}$ is an loose endo-arrow $A_X : X \multimap X$ equipped with *globular multiplication* and **unit** cells

$$\begin{array}{ccc} X & \xrightarrow{A} X & \xrightarrow{A} X \\ \parallel & \Downarrow \mu_A & \parallel \\ X & \xrightarrow[A]{} X & \end{array} \quad \begin{array}{ccc} X & \xrightarrow{1_X} X & \\ \parallel & \Downarrow \eta_A & \parallel \\ X & \xrightarrow[A]{} X & \end{array}$$

satisfying the usual associativity and unit laws. Given two monads A_X, B_Y , a **monad morphism** is a cell $f\alpha^f \equiv \alpha_f : A_X \Rightarrow B_Y$ that respects the multiplication and the units:

$$\begin{array}{ccccc} \begin{array}{ccccc} X & \xrightarrow{A} X & \xrightarrow{A} X \\ f \downarrow & \alpha & f \downarrow & \alpha & \downarrow f \\ Y & \xrightarrow{B} Y & \xrightarrow{B} Y \\ \parallel & \Downarrow \mu_B & \parallel \\ Y & \xrightarrow[B]{} Y \end{array} & = & \begin{array}{ccccc} X & \xrightarrow{A} X & \xrightarrow{A} X \\ \parallel & \Downarrow \mu_A & \parallel \\ X & \xrightarrow[A]{} X \\ f \downarrow & \alpha & \downarrow f \\ Y & \xrightarrow[B]{} Y \end{array} & & \begin{array}{ccccc} X & \xrightarrow{1_X} X & \\ \parallel & \Downarrow \eta_A & \parallel \\ X & \xrightarrow[A]{} X \\ f \downarrow & \alpha & \downarrow f \\ Y & \xrightarrow[B]{} Y \end{array} & = & \begin{array}{ccccc} X & \xrightarrow{1_X} X & \\ f \downarrow & 1_f & \downarrow f \\ Y & \xrightarrow[1_Y]{} Y \\ \parallel & \Downarrow \eta_B & \parallel \\ Y & \xrightarrow[B]{} Y \end{array} \end{array}$$

Dually, we can define comonads and comonad morphisms.

Monads and monad morphisms assemble into a category, $\mathbf{Mnd}(\mathbb{D})$. Dually, comonads and comonad morphisms assemble into the category $\mathbf{Cmd}(\mathbb{D})$.

There are obvious forgetful functors

$$\mathbf{Mnd}(\mathbb{D}) \rightarrow \mathbf{End}(\mathbb{D}) \quad \text{and} \quad \mathbf{Cmd}(\mathbb{D}) \rightarrow \mathbf{End}(\mathbb{D}).$$

Then, under certain assumptions on $\mathbb{D}$,

- there is a (co)free (co)monad,
- (co)monads are (co)monadic over the loose endo-arrows as soon as the free (co)monad exists, and
- the category of (co)monads is locally presentable.

Examples. 1. $\text{Mnd}(\text{Span}(\mathcal{C})) = \text{Cat}(\mathcal{C})$, the category of categories internal to $\mathcal{C}$.

loose endo-arrow $A_{X_0} : X_0 \rightharpoonup X_0$	$\iff$	source and target $X_0 \xleftarrow{s} X_1 \xrightarrow{t} X_0$
multiplication $\mu_A : A \odot A \Rightarrow A$	$\iff$	composition $X_1 \times_{X_0} X_1 \rightarrow X_1$
unit $\eta_A : 1_A \Rightarrow A$	$\iff$	identity $X_0 \rightarrow X_1$
monad morphism	$\iff$	internal functor

On the other hand, $\text{Cmd}(\text{Span}(\mathcal{C}))$ are just arrows in $\mathcal{C}$.

loose endo-arrow $A_X : X \rightharpoonup X$	$\iff$	span $X \xleftarrow{f} Y \xrightarrow{g} X$
counit $\varepsilon_A : A \Rightarrow 1_A$	$\iff$	$f = g = \varepsilon_A$

2. $\text{Mnd}(\text{Rel}(\text{Set})) = \text{Preord}(\text{Set})$, the category of reflexive and transitive relations.

loose endo-arrow $R_X : X \rightharpoonup X$	$\iff$	relation $R \subseteq X \times X$
multiplication $\mu_R : R \odot R \Rightarrow R$	$\iff$	$RR \subseteq R \iff R$ is transitive
unit $\eta_R : 1_R \Rightarrow R$	$\iff$	$\Delta_X \subseteq R \iff R$ is reflexive
monad morphism	$\iff$	order preserving morphism

A comonad $C_X : X \rightarrow X$ in $\text{Cmd}(\text{Rel}(\text{Set}))$ is just a subobject of X .

$$\text{counit } \varepsilon_C : C \Rightarrow 1_C \iff C \subseteq \Delta_X$$

3. $\text{Mnd}(\mathbb{R}\text{ing}) = \text{Alg}$, the global category of algebras over rings.

loose endo-arrow $M_R : R \rightarrowtail R$	$\Leftrightarrow$	(R, R) -bimodule M
multiplication $\mu_M : M \odot M \Rightarrow M$	$\Leftrightarrow$	multiplication $M \otimes_R M \rightarrow M$
unit $\eta_M : 1_M \Rightarrow M$	$\Leftrightarrow$	unit $R \rightarrow M$
monad morphism	$\Leftrightarrow$	algebra homomorphism

Dually, $\text{Cmd}(\mathbb{R}\text{ing})$ is the global category Coalg of coalgebras over rings.

4. $\text{Mnd}(\mathcal{V}\text{-Mat}) = \mathcal{V}\text{-Cat}$.

loose endo-arrow $A_X : X \rightarrowtail X$	$\Leftrightarrow$	family of objects $A_{x,x}$ in $\mathcal{V}$
multiplication $\mu_A : A \odot A \Rightarrow A$	$\Leftrightarrow$	morphisms $A_{x',x''} \otimes A_{x,x'} \rightarrow A_{x,x''}$
unit $\eta_A : 1_A \Rightarrow A$	$\Leftrightarrow$	$I \rightarrow A_{x,x'}$
monad morphism	$\Leftrightarrow$	$\mathcal{V}$ -functors

Dually, $\text{Cmd}(\mathcal{V}\text{-Mat})$ are $\mathcal{V}$ -cocategories and $\mathcal{V}$ -cofunctors.

(co)Free (co)monads

Theorem (free monads, [ASTV])

Suppose that $\mathbb{D}$ is a fibrant double category. If either of the following holds

1. $\mathbb{D}_0$ and $\mathcal{H}(\mathbb{D})(X, Y)$ have coproducts which are preserved by $A \odot -$ and $- \odot A$ for any $X, Y \in \mathbb{D}_0$ and $A \in \mathbb{D}_1$.
2. $\mathbb{D}$ is locally presentable

then the forgetful functor $U: \text{Mnd}(\mathbb{D}) \rightarrow \text{End}(\mathbb{D})$ has a left adjoint.

Proof. Both $\text{Mnd}(\mathbb{D})$ and $\text{End}(\mathbb{D})$ are fibered over $\mathbb{D}_0$ [ASV] and U is a fibered functor between these fibrations

$$\begin{array}{ccc} \text{Mnd}(\mathbb{D}) & \xrightarrow{U} & \text{End}(\mathbb{D}) \\ & \searrow \quad \swarrow & \\ & \mathbb{D}_0 & \end{array}$$

So U has a left adjoint as long as each restriction on the fibers $U_X: \text{Mnd}(\mathbb{D})_X \rightarrow \text{End}(\mathbb{D})_X$ has a left adjoint. $\square$

Point (2) can be dualized.

(co)Monadicity

Theorem (monadicity of monads, [ASTV])

If $\mathbb{D}$ is a fibrant double category such that the forgetful functor $U : \text{Mnd}(\mathbb{D}) \rightarrow \text{End}(\mathbb{D})$ has a left adjoint then $\text{Mnd}(\mathbb{D})$ is monadic over $\text{End}(\mathbb{D})$.

Proof. By Beck's Monadicity Theorem. U has a left adjoint by hypothesis and reflects isos. Lastly, If a pair of parallel *monad* maps

$$\begin{array}{ccc} X & \xrightarrow{M} & X \\ f \downarrow & \alpha_f, \beta_g & \downarrow g \\ Y & \xrightarrow{N} & Y \end{array}$$

have a *split* coequalizer in $\text{End}(\mathbb{D})$

$$\begin{array}{ccccc} & T_t & & S_s & \\ & \curvearrowright & & \curvearrowright & \\ M_X & \xrightarrow{\alpha_f} & N_Y & \xrightarrow{\gamma_h} & C_Z \\ & \xleftarrow{\beta_g} & & & \end{array}$$

then C_Z can be given the structure of a monad such that the above diagram is a coequalizer in $\text{Mnd}(\mathbb{D})$.

For the multiplication apply $-\odot-$ to the previous diagram to obtain another coequalizer and use the multiplication on N .

$$\begin{array}{ccccc}
 M_X \odot M_X & \xrightarrow[\beta_g \odot \beta_g]{\alpha_f \odot \alpha_f} & N_Y \odot N_Y & \xrightarrow{\gamma_h \odot \gamma_h} & C_Z \odot C_Z \\
 \downarrow \mu_M & & \downarrow \mu_N & & \downarrow \exists! \mu_C \\
 M_X & \xrightarrow[\beta_g]{\alpha_f} & N_Y & \xrightarrow{\gamma_h} & C_Z
 \end{array}$$

For the unit apply $1_{(-)} \circ \mathfrak{s}$

$$\begin{array}{ccccc}
 1_X & \xrightarrow[1_g]{1_f} & 1_Y & \xrightarrow{1_h} & 1_Z \\
 \downarrow \eta_M & & \downarrow \eta_N & & \downarrow \exists! \eta_C \\
 M_X & \xrightarrow[\beta_g]{\alpha_f} & N_Y & \xrightarrow{\gamma_h} & C_Z
 \end{array}$$

and use similar arguments. □

Everything can be *dualized*, giving comonadicity of comonads over $\text{End}(\mathbb{D})$.

Local presentability

Theorem ($\mathbf{Mnd}(\mathbb{D})$ is locally presentable, [ASTV])

If $\mathbb{D}$ is locally presentable and the functors $-\odot-$ and $1_{(-)}$ preserve filtered colimits then $\mathbf{Mnd}(\mathbb{D})$ is locally presentable.

Proof. Consider the endofunctor

$$F: \mathbf{End}(\mathbb{D}) \rightarrow \mathbf{End}(\mathbb{D})$$

$$M_X \mapsto (M_X \odot M_X) + M_X.$$

F is finitary and $\mathbf{End}(\mathbb{D})$ is locally presentable since $\mathbb{D}$ is [ASV], therefore $\mathbf{Alg}(F)$ is locally presentable.

$\mathbf{Mnd}(\mathbb{D})$ is a full subcategory of $\mathbf{Alg}(F)$ closed under limits and filtered colimits. Thus, by the Reflection Theorem [AR], $\mathbf{Mnd}(\mathbb{D})$ is also locally presentable. $\square$

The above theorem *cannot* be dualized.

Theorem ($\text{Cmd}(\mathbb{D})$ is locally presentable, [ASTV])

Suppose $\mathbb{D}$ is locally presentable and the functors $- \odot -$ and $1_{(-)}$ preserve filtered colimits. If furthermore, the categories $\mathcal{H}(\mathbb{D})(X, Y)$ have initial objects preserved by $A \odot -$ and $- \odot A$ for any A , then $\text{Cmd}(\mathbb{D})$ is locally presentable.

Proof. Consider the endofunctor

$$F : \text{End}(\mathbb{D}) \rightarrow \text{End}(\mathbb{D})$$

$$C_X \mapsto (C_X \odot C_X) \times 1_X.$$

Again, F is finitary and $\text{End}(\mathbb{D})$ is locally presentable [ASV], therefore $\text{Coalg}(F)$ is also locally presentable.

$\text{Cmd}(\mathbb{D})$ is contained in $\text{Coalg}(F)$ but for F -coalgebras, their structure maps are neither globular nor do they need to satisfy any coassociativity and counitality conditions.

Consider first the full subcategory $\overline{\text{Coalg}(F)}$ of $\text{Coalg}(F)$ spanned by F -coalgebras with *globular* structure maps.

$\overline{\text{Coalg}(F)}$ is closed under all colimits in $\text{Coalg}(F)$, thus cocomplete.

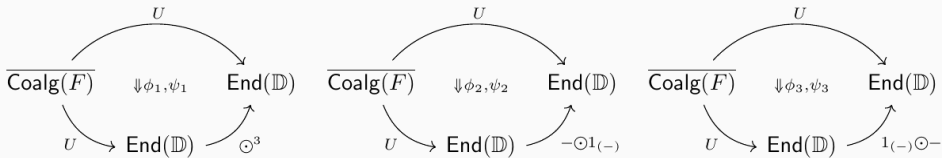
Furthermore, we can prove that

Lemma [ASTV]

Under the above assumptions if $\alpha_f : C_X \Rightarrow D_Y$ is a monomorphism in $\text{Coalg}(F)$ then its source/target $f : X \rightarrow Y$ is a monomorphism in $\mathbb{D}_0$.

and thus conclude that $\overline{\text{Coalg}(F)}$ is closed under all subobjects in $\text{Coalg}(F)$ hence accessible [AR]. So $\overline{\text{Coalg}(F)}$ is locally presentable.

$\text{Cmd}(\mathbb{D})$ is now the equifier of



in $\overline{\text{Coalg}(F)}$, that is the subcategory spanned of those objects C for which $(\phi_i)_C = (\psi_i)_C$ for $i = 1, 2, 3$ where

$$\left\{ \begin{array}{l} (\phi_1)_C = (\delta_C * \text{id}_C) \circ \delta_C : C \rightarrow (C \odot C) \odot C \\ (\psi_1)_C = (\text{id}_C * \delta_C) \circ \delta_C : C \rightarrow C \odot (C \odot C) \\ (\phi_2)_C = (\epsilon_C * \text{id}_C) \circ \delta : C \rightarrow 1_{s(C)} \odot C \\ (\psi_2)_C = \ell_C : C \rightarrow 1_{s(C)} \odot C \\ (\phi_3)_C = (\text{id}_C * \epsilon_C) \circ \delta : C \rightarrow C \odot 1_{t(C)} \\ (\psi_3)_C = r_C : C \rightarrow C \odot 1_{t(C)} \end{array} \right.$$

All functors are accessible, therefore $\text{Cmd}(\mathbb{D})$ is accessible. Furthermore it is cocomplete [ASV], hence locally presentable.

- [AR] J. Adámek, J. Rosický, *Locally presentable and Accessible Categories*, London Mathematical Society Lecture Note Series 189, Cambridge University Press.
- [ASV] V. Aravantinos-Sotiropoulos, C. Vasilakopoulou, *Sweedler Theory for double categories*. arXiv:2408.03180.
- [ASTV] V. Aravantinos-Sotiropoulos, T.T., C. Vasilakopoulou, *On categories of monads and comonads in double categories*. arXiv:2606.26755.

Thank you for your attention!